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Evidence for the existence of a magnetic quantum critical point in $U(Pt_{1-x}Pd_x)_3$

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Abstract

We report systematic studies of the DC magnetic susceptibility and electrical resistivity on a series of $U(Pt_{1-x}Pd_x)_3$ polycrystals with $0 \leqslant x \leqslant 0.02$. Susceptibility measurements show that the spin fluctuation temperature varies linearly with x. Combined with susceptibility data for UPt_3 under hydrostatic pressure and $(U,Th)Pt_3$ we find that T_{SF} is directly proportional to the c/a ratio in this (nearly) HCP-like system. Detailed studies of the resistivity show that increasing x results in a deviation from Fermi-liquid behavior, and the power-law exponent reaches a minimum value of 1.55 near $x \approx 0.006$. These results lend further support to the previously reported magnetic quantum critical point at $x \approx 0.006$ as determined via local-probe μSR measurements. © 2002 Elsevier Science B.V. All rights reserved.

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The heavy-fermion superconductor UPt_3 has been an intensely studied system for nearly two decades, and much of the seminal work on this material was carried out in the Franse research group at the Van der Waals–Zeeman Institute of the University of Amsterdam [1]. In this intriguing system, a complex magnetic fluctuation spectrum gives rise to a strongly correlated heavy fermion state at low temperatures. At 6 K, the onset of an anomalous antiferromagnetic phase is observed via elastic neutron scattering [2]. It is anomalous as the size of the ordered moment is quite small (roughly $0.02\,\mu_B$), the square of the ordered moment increases quasi-linearly with decreasing temperature, and all other bulk and local probe measurements (e.g., NMR [3] and μSR [4,5]) do not detect entry into this phase. Recent NMR measurements [6] on another small moment antiferromagnet, URu_2Si_2, reveal that the origin of its small moment, as detected by neutron scattering, results from phase-separated regions of paramagnetism and conventional antiferromagnetism, with the latter occupying a fairly small volume of the sample. However, to date no evidence exists for phase separation in UPt_3. It has been pointed out (see for example [7]) that the results for UPt_3 may result from a magnetic order which fluctuates on a time scale faster than that for muon and nuclear magnetic measurements (10^{-7} s), but slower than the characteristic time of neutron scattering (10^{-11} s). The nature of the small-moment antiferromagnetic state (or "SMAF") is quite intriguing, and the fact that it gives rise to an unconventional superconducting

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state below 0.6 K [8] makes its understanding an important unresolved question.

As described in detail by de Visser in this volume, neutron scattering and muon spin rotation studies of the evolution of magnetism in $U(Pt_{1-x}Pd_x)_3$ have proven to be a powerful probe of its relationship to superconductivity in this system. Substitution of small amounts of Pd for Pt suppresses superconductivity more rapidly than other impurities (both magnetic and non-magnetic), with complete suppression occurring at $x_{C,SC} = 0.006$ [9]. At higher concentrations ($x > 0.01$) neutron [10], transport, NMR [11], and μSR [4] measurements reveal the onset of what appears to be a conventional large-moment antiferromagnetic phase (LMAF). Remarkably, extrapolation of the phase line determined from μSR data with $0.007 \leqslant x \leqslant 0.010$ to 0 K indicates a magnetic quantum critical point $x_{C,AFM} = x_{C,SC} = 0.006$ [12]. Thus, while anomalous small-moment antiferromagnetism coexists with, and couples to, superconductivity, conventional large-moment antiferromagnetism appears to directly compete with it. This proposed relationship is in stark contrast to the behavior recently observed in $CeIn_3$ and $CePd_2Si_2$ [13], where superconductivity is stabilized in the vicinity of an antiferromagnetic quantum critical point.

A key point in understanding the relationship between the SMAF, LMAF, and superconducting states is whether or not the extrapolation presented in Ref. [12] is correct, and a true magnetic quantum critical point exists at $x_{C,AFM} = 0.006$. In this paper we will present some recent bulk electronic transport and magnetic susceptibility measurements on $U(Pt_{1-x}Pd_x)_3$ polycrystals with $0 \leqslant x \leqslant 0.02$. The objective of this work is to provide complementary measurements to local probe μSR measurements in Ref. [12] in order to clarify the existence of a magnetic quantum critical point at $x = 0.006$. Specifically, we look to determine if (a) there is an interaction parameter that is smoothly varied by the addition of Pd, and (b) to determine if any bulk evidence of non-Fermi liquid behavior exists in the vicinity of the proposed magnetic quantum critical point.

In an earlier work [14] the temperature dependent resistivity of $U(Pt_{1-x}Pd_x)_3$ was re-analyzed for several samples fabricated in Amsterdam over the years. While the results indicated that as the Pd concentration was increased the temperature dependence deviated from the quadratic behavior expected for Fermi liquids, there was significant scatter in those results, presumably due to differing amounts of unwanted impurities, variations in annealing procedures, etc. Thus, to minimize these differences, a single series of $U(Pt_{1-x}Pd_x)_3$ polycrystals was manufactured. The series ($x = 0$, 0.004, 0.005, 0.006, 0.007, 0.008, 0.010, 0.014, 0.020) was made by first creating master alloys (0 and 2 at% Pd), then using these to produce samples with $0 < x < 0.02$). At a later date samples with $x = 0.012$ and 0.013 were made under the same conditions. For starting materials we used U (99.98%), Pt (99.99%), and Pd (99.999%). Stoichiometric amounts of the starting materials, or of the master alloys, were melted in an arc furnace on a water-cooled Cu crucible in 0.5 bar of ultrapure argon gas. The resulting buttons were wrapped in tantalum foil and put in water free quartz tubes together with a piece of uranium to serve as a getter. After evacuating ($p < 10^{-6}$ mbar) and sealing the tubes, the samples were annealed at 950°C for 5 days and gradually cooled over 4 days to 25°C. The annealed samples were cut into bars via the spark erosion process.

The sample magnetization was measured in a 4He gas flow cryostat using the DC extraction technique in an applied magnetic field of 6 T. Since we observe the magnetization to vary linearly with applied field up to 6 T for temperatures $2\,K \leqslant T \leqslant 150\,K$, the magnetic susceptibility was obtained by the ratio of magnetization to applied magnetic field. Typical data below 60 K for three samples are shown in Fig. 1. The maximum below 20 K is associated with the stabilization of magnetic spin fluctuations [15], and the temperature at which the maximum occurs is the spin fluctuation temperature T_{SF}. While the maximum in the susceptibility occurs at roughly 11 K for $x = 0.02$, the onset of antiferromagnetic order occurs at $T_N = 4\,K$, as determined by a variety of local and bulk probes, and is signaled by a break in the slope of susceptibility versus temperature (see Fig. 2).

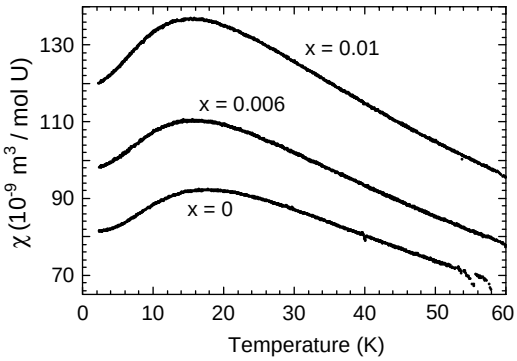

Fig. 1. Magnetic susceptibility, as calculated from the magnetization in a field of 6 T, for three polycrystalline samples of $U(Pt_{1-x}Pd_x)_3$.

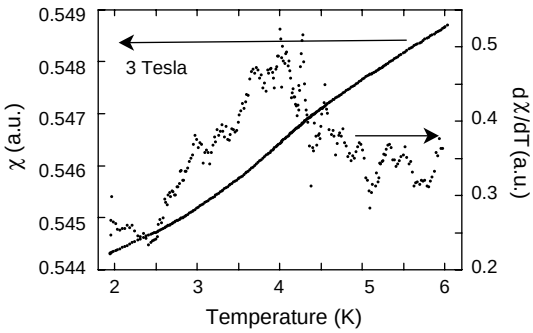

Fig. 2. Magnetic susceptibility, measured using a cantilever magnetometer, and its temperature derivative, as a function of temperature in the vicinity of the Néel temperature $T_N = 4.0$ K (determined independently from resistivity measurements) for a sample with $x = 0.02$.

We find that T_{SF} decreases linearly with increasing Pd concentration. Our detailed results are consistent with de Visser's earlier results [16], which were taken for samples with higher values of x. In that work, it was speculated that the decrease in T_{SF} corresponded to an increase in the low temperature spin fluctuations. This in turn was speculated to arise from the systematic reduction of the c/a ratio in this nearly HCP-like system.[1] To test this hypothesis, we have plotted in Fig. 3 the spin-fluctuation temperature versus the fractional change in the c/a ratio relative to pure UPt_3 for

[1] It has recently been observed that the crystal structure has very slight c-axis displacements, and the actual crystal symmetry group is trigonal. See Ref. [17].

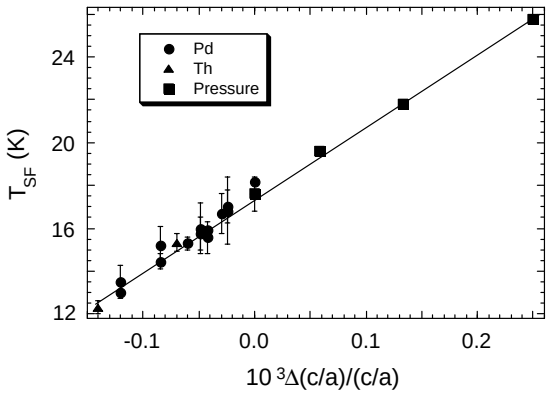

Fig. 3. Spin-fluctuation temperature T_{SF} as a function of the calculated fractional change in the c/a ratio (see text). Implicit variables are Pd and Th concentration and hydrostatic pressure. The solid line is a guide to the eye.

our data. The change in c/a for our samples was estimated using earlier X-ray data on samples with larger Pd-concentrations [16]. Also shown in the graph are previously reported data for UPt_3 under hydrostatic pressure [15], and for two samples of $U_{1-x}Th_xPt_3$ with $x = 0.01$ and 0.02 (samples fabricated by J.C. Cooley and J.L. Smith [18]). The changes in the c/a ratio due to pressure and Th-substitution were calculated from data presented in Refs. [16,19], respectively. We see that all the data fall on a straight line, strongly supporting the conjecture in Refs. [16,20]: the magnetic interactions in UPt_3 are governed by c/a.

The temperature-dependent electrical resistivity was also measured for these samples via an AC 4-terminal technique at temperatures between 0.3 and 300 K, utilizing both a 3He refrigerator and a 4He gas flow cryostat. It has been previously reported that UPt_3 exhibits Fermi liquid-like resistivity below about 2 K, with

$$\rho = \rho_0 + AT^{\alpha}, \tag{1}$$

$\alpha = 2$. In fact α is typically somewhat smaller than, but still close to, 2 [14,21]. Fitting our data below 1 K to the generalized power law form given above, we find that α is systematically decreased with increasing x, reaching a minimum value of about 1.55 near $x = 0.007$. This is followed by a small increase, and at higher x α appears to approach a constant value near 1.65. This general

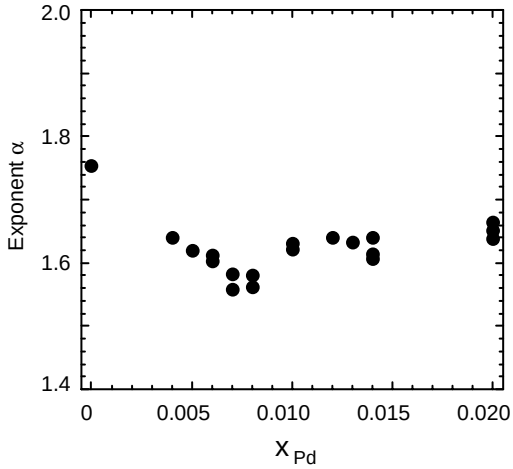

Fig. 4. Power law exponent of the resistivity as a function of Pd concentration from fitting the temperature-dependent resistivity data below 1 K to Eq. (1). Error bars due to the fitting procedure are smaller than the data point symbols. Multiple points at a given concentration correspond to different samples cut from the same button.

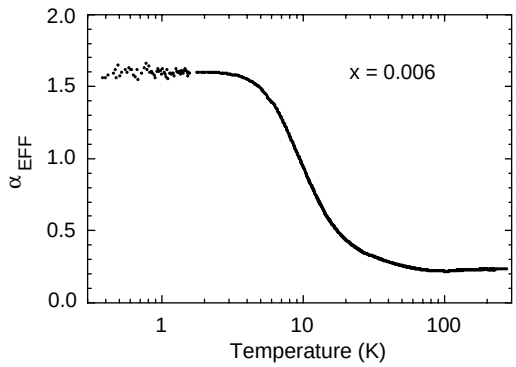

Fig. 5. Effective exponent variation with temperature, as calculated from the data using Eq. (2), for a sample with $x = 0.006$. The high temperature data were taken in a 4He gas flow cryostat; the larger scatter in the data below 1.5 K results from the reduced excitation currents required for the "sample-in-vacuum" 3He refrigerator.

behavior, shown in Fig. 4, is consistent with non-Fermi liquid behavior in the vicinity of the proposed magnetic quantum critical point. In Fig. 5 we show our data for the "effective exponent" for $x = 0.006$

$$\alpha_{\mathrm{EFF}}(T) \equiv \frac{\mathrm{d}\ln(\rho - \rho_0)}{\mathrm{d}\ln T}. \tag{2}$$

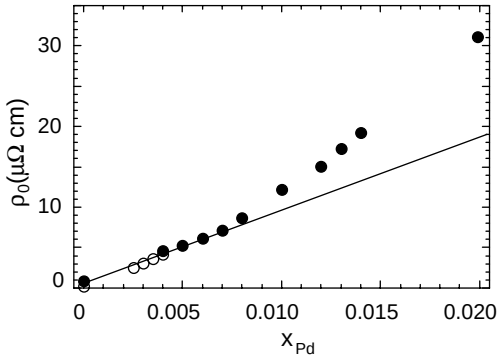

Fig. 6. Variation of the residual resistivity with Pd concentration, as determined by fitting the data below 1 K to Eq. (1). The solid line is a guide to the eye. Filled circles are the data presented in this work, while the open circles were measured for similarly prepared samples discussed in Ref. [14].

α_{EFF} is constant below about 3 K, indicating that the system exhibits true power law behavior at low temperatures.

The variation of the residual resistivity with x for these samples, along with data for similarly prepared samples prepared at lower x [14], is shown in Fig. 6. A strong deviation from linear behavior is observed at $x = 0.006$. As reported in many earlier works,[2] an increase in ρ_0 is often observed near the onset of magnetic ordering.

These data strongly suggest that substitution of Pd for Pt is a useful way to vary the chemical pressure in this system, producing opposite effects of hydrostatic pressure. However, Pd substitution produces disorder effects which cannot be ignored. We have found that for poor quality samples of *pure* UPt_3 with very high residual resistivities, α can be significantly lower than 2 (in one case as low as 1.4) [23]. A disorder-driven variation in α would be consistent with theoretical predictions for spin-fluctuating systems [24], and also experimental observations in disordered spin systems.[3] Still, the fact that α exhibits a minimum near x_C near the expected value of $\frac{3}{2}$ [26] strongly suggests the existence of non-Fermi liquid behavior in the vicinity of an antiferromagnetic quantum critical point. We are currently separating disorder effects

[2] For example, in MnSi under pressure as reported by Thessieu et al. [22].

[3] For example, as reported by Ford and Mydosh [25].

from changes in the power law exponent α caused by a variation of c/a by applying hydrostatic pressure to the sample with $x = 0.02$ in order to restore nearly Fermi liquid behavior as c/a is increased.

Our future measurements include a detailed search for evidence of non-Fermi liquid behavior in the vicinity of x_C from the thermodynamic properties. These include specific heat measurements, and the use of sensitive cantilever magnetometry to try and follow T_N towards 0 K as well as monitoring the temperature dependence of the susceptibility.

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